

Boiler Interlock Safety

Interlock Safety & Efficiency Overview

Purpose of This Table:

This table provides a reference guide for critical interlock components used in boiler systems to maintain safe, efficient, and compliant operations.

Each interlock listed plays a specific role in preventing equipment failure, reducing energy waste, and protecting both personnel and infrastructure. Understanding the purpose, impact on efficiency, and required maintenance of each device helps facilities:

- ⚠️ Prevent accidents caused by overheating, flame failure, dry-firing, or flow blockages
- 🔧 Improve system longevity by reducing wear-and-tear through smart shutdowns and controls
- 💰 Enhance operational efficiency by minimizing downtime, fuel waste, and erratic performance
- 📋 Stay compliant with OSHA, ASME, NFPA, and local inspection codes

Interlock	Purpose	Efficiency Impact	Maintenance Tip
Low Water Cutoff (LWCO)	Shuts off burner if water level is too low	Prevents dry-firing, extending life	Test weekly
High Limit Aquastat	Cuts off burner at max temp	Prevents overheating & scale buildup	Verify settings quarterly
Flame Safeguard System	Monitors flame integrity & startup sequence	Ensures clean ignition; reduces restarts	Test sensors & relays monthly
Pressuretrol or Modulating Controller	Controls firing rate based on pressure	Maintains steady loads, prevents short cycling	Calibrate annually
Flow Switch (Chilled or Condensate Return)	Verifies water or steam flow	Prevents overheating from pump or flow failures	Inspect paddle or probe quarterly
Stack Temperature Sensor	Shuts burner if temp exceeds limit	Avoids wasted fuel from poor heat transfer	Log trends with sensor if available
E-Stop Safety Switch	Emergency system shutdown	Protects all systems and staff	Ensure accessibility & test monthly